Team Number: 104
Project Name: Slip Detection Subsystem
Team Member Names: Aakash Kumar
Version: 1.2

A. Major Components

All Major Components	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Microcontroller	PIC18F57Q43	1.8V - 5.5V	1	50	50	mA
	Accelerometer	ADXL335BCPZ	1.8V - 3.6V	1	1	1	mA
	Opamp	TLV9062	1.8V - 5.5V	3	10	30	mA

B. Power Rails

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Accelerometer	ADXL335BCPZ	1.8V - 3.6V	1	1	1	mA
	Subtotal					1	mA
	Safety Margin					0.25	
	Total Current Required on +3.3V Rail					1.25	mA
c1. Regulator or Source Choice	**+3.3V Regulator**	**LD1117V33**	**4.3V - 15V**	**1**	**800**	**800**	**mA**
	Total Remaining Current Available on +3.3V Rail					798.75	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Microcontroller	PIC18F57Q43	1.8V - 5.5V	1	50	50	mA
	Opamp	TLV9062	1.8V - 5.5V	3	10	30	mA
	+3.3V Regulator Load	LD1117V33	4.3V - 15V	1	1.25	1.25	mA
	Subtotal					81.25	mA
	Safety Margin					0.25	
	Total Current Required on +5V Rail					101.56	mA
c2. Regulator or Source Choice	**+5V Regulator**	**L7805**	**7V - 35V**	**1**	**1000**	**1000**	**mA**
	Total Remaining Current Available on +5V Rail					898.44	mA

C. Confirm Voltage Regulators

D. External Power Source

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	**9V Battery via Barrel Jack**	**4166-PART9VBATTERYBARRELCONNECTOR-ND**	**+9V**	**+9V**	**500**	**500**	**mA**
Power Rails Connected to External Power Source 1	**+5V Regulator**	**L7805**	**7V - 35V**		**101.56**	**101.56**	**mA**
	Total Remaining Current Available on External Power Source 1					300	mA

E. Calculate Battery Life

Component Name	Part Number	Supply Voltage Range	Capacity (mAh)	Required By Regulators	
9V Battery via Barrel Jack	4166-PART9VBATTERYBARRELCONNECTOR-ND	+9V	500	101.56	
			Battery Life	4.92	hours